

The Waterfall Model in Large-Scale Development

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Abstract. Waterfall development is still a widely used way of working in software development companies. Many problems have been reported related to the model. Commonly accepted problems are for example to cope with change and that defects all too often are detected too late in the software development process. However, many of the problems mentioned in literature are based on beliefs and experiences, and not on empirical evidence. To address this research gap, we compare the problems in literature with the results of a case study at Ericsson AB in Sweden, investigating issues in the waterfall model. The case study aims at validating or contradicting the beliefs of what the problems are in waterfall development through empirical research.

1 Introduction

The first publication on the waterfall model is credited to Walter Royce's article in 1970 (cf. [1]). In literature there seems to be an agreement on problems connected to the use of the waterfall model. Problems are (among others) that the model does not cope well with change, generates a lot of rework, and leads to unpredictable software quality due to late testing [2]. Despite the problems identified, the model is still widely used in software industry, some researchers are even convinced that it will be around for a much longer period of time (see [3]). The following trends can be seen in research. First, the model seems to be of very little interest for researchers to focus on as it seems to be old-fashioned. Instead, recent studies have much more focus on agile and incremental development. Secondly, there is very little empirical research backing up what we believe to know about the waterfall model. In order to identify the evidence provided by empirical research on the waterfall model we conducted the following search on Inspec & Compendex:

- (“waterfall model” OR “waterfall development”) AND (“empirical” OR “case study” OR “industrial”)

Inspec & Compendex was selected as it integrates many full-text databases in computing and thus is considered a good starting point. The search resulted in 33 publications where none of the publications had an explicit focus on studying the waterfall model in an industrial setting. Thus, most of the problems reported on the waterfall model are mainly based on researchers' beliefs and experience reports. Consequently, in order to provide substantial evidence on the usefulness of the waterfall model in industry empirical studies are needed. Evaluating the usefulness empirically aids decision making of whether to use the model in specific context (here large-scale-development).

To address this research gap we conducted a case study focusing on identifying issues in waterfall development and compare them to what has been said in literature. Furthermore, the issues identified are ranked based on their criticality. The case being studied is a development site of Ericsson AB, Sweden. The waterfall model was used at the company for several years. The case study has been conducted according to the guidelines provided by Yin (see [4]). The case study makes the following contributions to research on waterfall development: 1) Illustration of the waterfall implementation in practice within large-scale industrial software development, 2) Identification of issues related to the waterfall model and their prioritization showing the most critical issues, and 3) Comparison of case study results with state of the art (SotA).

The remainder of this paper is structured as follows: Section 2 provides an overview of related work. Thereafter, Section 3 illustrates the waterfall model used at the company. Section 4 presents the case study design. The analysis of the collected data is provided in Section 5 (qualitative analysis) and Section 6 (quantitative analysis). Section 7 presents a comparison of the case study findings and state of the art. Section 8 concludes the paper.

2 Related Work

Literature identifies a number of problems related to the waterfall model. An overview of the problems identified in literature is shown in Table 1. In addition to the identified articles we considered books discussing advantages and disadvantages of the waterfall model.

The waterfall model is connected to high costs and efforts [2][5]. That is, it requires approval of many documents, changes are costly to implement, iterations take a lot of effort and rework, and problems are usually pushed to later phases [2]. Few studies are explicitly focused on the waterfall model and some reasons for the failures of the waterfall approach have been identified. One reason mentioned by several studies is the management of a large scope, i.e. requirements cannot be managed well and has been identified as the main reason for failure (cf. [7] [9] [8]). Consequences have been that the customers' current needs are not addressed by the end of the project [7], resulting in that many of the features implemented are not used [9].

Additionally, there is a problem in integrating the overall system in the end and testing it [10]. A survey of 400 waterfall projects has shown that the software being developed is either not deployed or if deployed, it is not used. The